

Athanasios Angelakis

Curriculum Vitae

Senior Biomedical AI Researcher specializing in trustworthy machine learning, multimodal biomedical data science, explainable AI, and AI/statistical methodology for clinical research.

Selected Methodological Strengths

AI/ML	Explainable AI, multimodal learning, robust low-data learning, feature selection, data-centric AI, synthetic data generation, foundation models, LLMs, clinical decision-support systems.
Statistics	Model validation, survival analysis, bootstrap confidence intervals, bias and confounding assessment, high-dimensional biomedical data analysis, reproducibility.
Biomedical data	Omics, clinical tabular data, ultrasound, medical imaging, questionnaire data, text, multi-modal biomedical datasets.
Implementation	Python, PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, HuggingFace, LangChain, PySpark, Linux/HPC-style workflows.

Work Experience

- 02.2026–
present **Postdoctoral Researcher, AI & Biometrics Security, BioML Lab, CODE Research Institute, Universität der Bundeswehr München, Munich, Germany.**
Leading research on privacy-preserving biometric recognition systems, with emphasis on ECG-based authentication, adversarially robust machine learning, explainable AI, and secure multimodal biometric verification. My work extends my broader research on trustworthy and data-centric AI, including data augmentation methods for tabular and multimodal biomedical datasets, bias-aware learning, robust representation learning, and reproducible evaluation under distribution shift. I also supervise two PhD students working on face recognition and fully homomorphic encryption for biometric systems.
- 11.2025–
present **Affiliated Researcher / Sr. AI Data Scientist, Amsterdam UMC, Amsterdam, NL.**
Continuing research on data-centric and explainable artificial intelligence, focusing on feature selection, bias-aware learning, and advanced data augmentation methodologies for tabular, imaging, and multimodal biomedical datasets. My work integrates multimodal machine learning pipelines combining clinical, imaging, anthropometric, and -omics data for diagnosis, prognosis, and biomarker discovery, with emphasis on robust model generalization, reproducibility, and trustworthy AI deployment in clinical decision-support systems.

- 11.2022–
10.2025 **Sr. Research AI Data Scientist**, AMSTERDAM UNIVERSITY MEDICAL CENTER, AMSTERDAM PUBLIC HEALTH RESEARCH INSTITUTE, UNIVERSITY OF AMSTERDAM - DEPARTMENT: EPIDEMIOLOGY & DATA SCIENCE, Amsterdam, Netherlands.
Conducting research on explainable artificial intelligence, feature selection, data augmentation, and bias across diverse data modalities, including tabular, image, video, text, and multimodal datasets. My work integrates multimodal AI pipelines that combine imaging, clinical, anthropometric, and -omics data for diagnosis, prognosis, and outcome prediction. Examples include: detecting pulmonary edema from ultrasound videos; classifying fibrosis and steatosis stages in chronic liver disease, with analysis encompassing over 20,000 biopsy-validated patients across multiple data modalities; identifying acute myeloid leukemia from -omics data; and integrating proteomics, transcriptomics, and imaging for biomarker discovery. I have also developed multimodal learning architectures that fuse deep learning, gradient boosting, and large language model (LLM) embeddings for clinical decision support. I introduced the concept of the Machine Learning Enhanced Non-Invasive Test, presented at the EASL International Liver Congress and AASLD Congress. Additionally, I have contributed to generative AI applications in biomedicine, such as synthetic data generation for rare disease cohorts, prompt-engineered LLMs for adverse drug reaction management, and LLM-based multimodal retrieval-augmented generation systems for biomedical knowledge extraction.
- 01.2022–
06.2022 **Machine/Deep Learning Consultant**, MASSACHUSETTS GENERAL HOSPITAL-HARVARD MEDICAL SCHOOL, MA, USA.
Research and development on Artificial Intelligence in Ultrasound, combining B-mode imaging with clinical and demographic data in multimodal frameworks. Developed algorithms for classification and segmentation, and applied contrastive learning to leverage large-scale unlabeled datasets. Incorporated generative models for ultrasound data augmentation, improving downstream classification accuracy.
- 03.2019–
12.2021 **Machine/Deep Learning Consultant**, HOLOGIC-SUPERSONIC IMAGINE, Aix-en-Provence, France.
Project leader on Prediction of Malignancy Risk of Breast Lesions using Ultrasound parameters. In particular, using B-mode measurements, the B-mode score and shear wave elastography measurements applying machine learning we were able to identify if a breast lesion was benign or malignant. Publishing 3 papers at ECR2020 and AIUM2021.
- 08.2020–
05.2021 **Sr. Data Scientist/Deep Learning Researcher**, HUAWEI, AMSTERDAM RESEARCH CENTER, Amsterdam, Netherlands.
Working on projects related to maps and computer vision using machine and deep learning. In particular, being member of the computer vision team I was leader of a detection and blurring of faces and vehicle license plates in videos recorded from cellphones project. Being a member of the routing team I was leader of 5 data imputation machine learning projects regarding different speed profiles in 25 countries. For all the above 6 projects I was responsible for the analysis, timelines, deliverables and tasks' management of them. The solutions were successfully delivered to head quarters and put in production. I was leading also a project which had to do with point-to-point ETA using deep learning. The proof of concept performance of the model for 25 countries was $MAPE \leq 0.02$. I have extensive experience analyzing OSM and TomTom's data regarding different speed profiles in 28 countries.
- 11.2018–
07.2020 **Lead Data Scientist**, PLOTWISE, Delft, Netherlands.
Delivered three Data Science projects in production applying ML, DL and RL methods on Route Planning, Traveling Salesman Problem, Geocoding, Shortest Path Distance, Traffic, Clustering. All had positive impact to Route Quality. I was communicating directly to the CEO, COO & CTO leading a team of 5-7 members, depending on the projects, consisted of data scientists, data analysts, data engineer, GIS engineer and software engineers.
- 06.2018–
10.2018 **Sr. Data Scientist/Algorithm Engineer**, PLOTWISE, Delft, Netherlands.
I created a Simulator for Dynamic Route Planning. Also I applied Deep Learning methods and Reinforcement Learning on Travelling Salesman Problem, Geocoding and Shortest Path Distance which all had positive impact to better Route Quality. I worked also in Route Optimization.

- 04.2017– **Sr. Data Scientist**, DIAGNOSTIC ECHOTOMOGRAPHY S.A., Athens, Greece.
 05.2018 I used Machine/Deep Learning algorithms on numeric and image data sets produced from Ultrasound modality. I was leading projects such as: full classification of liver Fibrosis and Steatosis, communicating directly to the CEO. Leading a team of 8-12 members (hepatologists, ultrasound examiners, pathologists, medical physicists, software engineers, IT). The models were used as decision support tools during the creation of medical reports. Corresponding papers were presented as Scientific Research Papers in conferences like ECR, ACR, RSNA, AIUM.
- 12.2016– **Post-Doc Researcher**, NATIONAL TECHNICAL UNIVERSITY OF ATHENS, Athens, Greece.
 12.2018 Research on projects related to Computational Number Theory and Machine/Deep Learning Algorithms.
- 01.2016– **Data Scientist**, *Freelancer*, Athens, Greece.
 01.2019 I have worked either as a Freelancer or on Academic Projects on Creation of Machine Learning Models with various kind of Numeric, Nominal or Image Data (Telecommunications, Genes, Energy, Health).
- 01.2016– **Post-Doc Researcher**, UNIVERSITY OF ATHENS, Athens, Greece.
 06.2016 Management and Research on Horizon2020 Project: Panoramix (MixNets and Lattice Based Cryptography).
- 10.2015– **Post-Doc Researcher**, MAX PLANCK INSTITUTE FOR MATHEMATICS, Bonn, Germany.
 12.2015 Research on Computational-Algorithmic Algebraic Number Theory.
- 10.2013– **Sr. Data Scientist/Algorithm Engineer**, ENOMIX, Athens, Greece.
 07.2015 Project Leader for the creation of a Next Generation Network (FTTx) Telecommunications software. Defining equal mass clusterization algorithms for urban, rural areas. Cost optimization in minimum spanning tree road networks, etc.. (Analysis, Design, Implementation of modules and algorithms). Communicating directly to the CEO and CTO, leading a team of 3 GIS/software Engineers. The software was used for consulting purposes by the company.

Teaching

- 06.2020– **Adjunct Professor**, EINDHOVEN UNIVERSITY OF TECHNOLOGY-JHERONIMUS ACADEMY
 12.2022 OF DATA SCIENCE, s'Hertogenbosch, Netherlands.
 Teaching and developing the courses “Deep Learning” and “Data Structures and Algorithms” in the Master's programme “Data Science and Entrepreneurship”. Conducting research on machine learning and deep learning, and supervising Master's students.
- 2018 Spring **Lecturer**, UNIVERSITY OF ATHENS, Athens, Greece.
 Semester Number Theory, Undergraduate course at the Department of Mathematics (in Greek).
- 2017 Spring **Lecturer**, UNIVERSITY OF ATHENS, Athens, Greece.
 Semester Number Theory, Undergraduate course at the Department of Mathematics (in Greek).
- 2012 Fall **Teaching Assistant**, UNIVERSITEIT LEIDEN, Leiden, Netherlands.
 Semester TA of Algebraic Number Theory, Dutch Master course in Mathematics (in English).
- 2011 Spring **Teaching Assistant**, UNIVERSITEIT LEIDEN, Leiden, Netherlands.
 Semester TA of Algebra I course, Universiteit Leiden (in Dutch).
- 09.2001– **Teaching Assistant**, UNIVERSITY OF ATHENS, Athens, Greece.
 08.2003 TA for Matlab in the Laboratories of Mathematics Department.
- 10.1999– **Mathematics Tutor**, SELF-EMPLOYED, Athens, Greece.
 06.2010 Private tutor of Mathematics to high school and university students.

Education

- 10.2010– **Ph.D. in Mathematics**, *Universiteit Leiden, Université de Bordeaux*, Leiden, Bordeaux.
 09.2015 Computational-Algorithmic Algebraic Number Theory
- 10.2008– **M.Sc. in Applied Mathematics**, *National Technical University of Athens*, Athens.
 06.2010 Specialized in Computational Mathematics

10.1998– **B.Sc. In Mathematics**, *National–Kapodistrian University of Athens*, Athens.
09.2006 Direction of Applied Mathematics, Specialized in Computational Mathematics

Open Source

<https://github.com/Bluesman79/ZACH-ViT>
pip install zachvit

Reviewer

1. Access, IEEE, Q1, H-Index: 290
2. International Journal of Computer Vision, Springer-Netherlands, Q1, H-Index: 232
3. Knowledge Based Systems, Elsevier B.V., Q1, H-Index: 188
4. Journal of Hepatology, Elsevier B.V., Q1, H-Index: 319
5. Computational and Structural Biotechnology, Elsevier B.V., Q1, H-Index: 92
6. Nutrition and Metabolism, BioMed Central Ltd, Q1, H-Index: 106
7. Journal of Medical Internet Research, JMIR Publications, Q1, H-Index: 214

Publications/Submissions

1. *Machine-Learning-Enhanced Non-Invasive Testing for MASLD Fibrosis: Shallow-Deep Neural Networks Versus FIB-4, Tabular Foundation Models, and Large Language Models*, A. Angelakis, G. De Vito, E.-M. Trifylli, F. Ferrucci, arXiv:2605.20523, <https://arxiv.org/abs/2605.20523>. (2026)
2. *Extending ZACH-ViT to Robust Medical Imaging: Corruption and Adversarial Stress Testing in Low-Data Regimes*, A. Angelakis, M. Gomez Barrero, <https://arxiv.org/abs/2604.06099>, CVPR Workshops (CVPRW), PHAROS Workshop on AI for Medical Imaging in Healthcare (AIF-MIH), (accepted) (2026)
3. *ZACH-ViT: Regime-Dependent Inductive Bias in Compact Vision Transformers for Medical Imaging*, A. Angelakis, arXiv:2602.17929, <https://doi.org/10.48550/arXiv.2602.17929>. (2026)
4. *A selective machine learning algorithm for severe periodontitis labeling from questionnaire data*, E. Stamatelou, N. Nijland, N. Su, B. G. Loos, A. Angelakis, Sci Rep (2026). <https://doi.org/10.1038/s41598-026-43934-6>
5. *LLMs for Drug–Drug Interaction Prediction Using Textual Drug Descriptors*, G. De Vito, F. Ferrucci, A. Angelakis, Knowledge-Based Systems, Elsevier, Article 115486, ISSN 0950-7051, <https://doi.org/10.1016/j.knosys.2026.115486>. (2026)
6. *Exploring the Role of Extracellular Vesicles in Pancreatic and Hepatobiliary Cancers: Advances Through Artificial Intelligence*, E. M. Trifylli, A. Angelakis, S. P. Fortis, A. G. Kriebardis, N. Papadopoulos, E. Koustas, P. Sarantis, M. V. Karamouzis, S. Manolakopoulos, M. Deutsch, International Journal of Molecular Sciences, MDPI, 27(3):1524, ISSN 1422-0067, <https://doi.org/10.3390/ijms27031524>. (2026)
7. *NAFLD Liver Fat Score Outperforms Controlled Attenuation Parameter for Diagnosing Hepatic Steatosis in Type 2 Diabetes: An Explainable Artificial Intelligence Study*, A. Angelakis, B. Lykiardopoulos, W. Balkhed, M. Bergram, F. Iredahl, M. Holmberg, C. Edin, C.-J. Carlhäll, T. Ebbens, P. Henriksson, C. Simonsson, K. Rådholm, G. Cedersund, M. Forsgren, O. Dahlqvist Leinhard, C. Jönsson, P. Lundberg, S. Kechagias, N. Dahlström, P. Nasr, M. Ekstedt, EASL ILC 2026 (accepted) (2026)

8. *AI-Based Fibrosis Detection in MASLD: Evaluating shallow deep neural networks, large language models and foundation models vs. FIB-4*, G. De Vito, E.-M. Trifylli, F. Ferrucci, A. Angelakis, EASL ILC 2026 (accepted) (2026)
9. *Circulating extracellular vesicles and ferritin enable machine learning based detection of hepatic steatosis in metabolic dysfunction associated steatotic liver disease*, E.-M. Trifylli, A. Angelakis, A. G. Kriebardis, N. Papadopoulos, S. Fortis, V. Pantazatou, I. Koskinas, H. Kranidioti, S. Manolakopoulos, M. Deutsch, EASL ILC 2026 (accepted) (2026)
10. *ZACH-ViT: A Zero-Token Vision Transformer with ShuffleStrides Data Augmentation for Robust Lung Ultrasound Classification*, Angelakis A. et al., arXiv:2510.17650, <https://doi.org/10.48550/arXiv.2510.17650>. (2025)
11. *HELIOT: LLM-Based CDSS for Adverse Drug Reaction Management*, G. De Vito, F. Ferrucci, A. Angelakis, Knowledge Based Systems, Elsevier, 114184, ISSN 0950-7051, <https://doi.org/10.1016/j.knosys.2025.114184>. (2025)
12. *Extracellular vesicles as MASLD biomarkers using explainable artificial intelligence: An algorithmic approach for steatosis staging*, Trifylli EM, Angelakis A, Kriebardis AG, Papadopoulos N, Fortis SP, Pantazatou V, Koskinas J, Kranidioti H, Koustas E, Sarantis P, Manolakopoulos S, Deutsch M. Extracellular vesicles as biomarkers for metabolic dysfunction-associated steatotic liver disease staging using explainable artificial intelligence. World J Gastroenterol 2025; 31(22): 106937 [DOI: 10.3748/wjg.v31.i22.106937]
13. *NAFLD Liver Fat Score Outperforms Controlled Attenuation Parameter for Diagnosing Hepatic Steatosis in Type 2 Diabetes: An Explainable Artificial Intelligence Study*, Athanasios Angelakis, Byron Lykiardopoulos, Wile Balkhed, Martin Bergram, Fredrik Iredahl, Markus Holmberg, Carl Edin, Carl-Johan Carlhäll, Tino Ebbers, Pontus Henriksson, Christian Simonsson, Karin Rådholm, Gunnar Cedersund, Mikael Forsgren, Olof Dahlqvist Leinhard, Cecilia Jönsson, Peter Lundberg, Mattias Ekstedt, Nils Dahlström, Patrik Nasr, Stergios Kechagias (submitted AASLD) (2025)
14. *AI-Based Fibrosis Detection in MASLD: Evaluating shallow-deep neural networks, large language models and foundation models vs. FIB-4*, Gabriele De Vito, Filomena Ferrucci, Athanasios Angelakis (submitted EASL) (2025)
15. *LLMs for Drug-Drug Interaction Prediction: A Comprehensive Comparison*, Gabriele De Vito, Filomena Ferrucci, Athanasios Angelakis (arXiv preprint arXiv:2502.06890) (2025)
16. *A shallow-deep neural network approach combining non-invasive tests to enhance advanced fibrosis detection in metabolic dysfunction-associated steatotic liver disease patients*, A. Angelakis (accepted EASL) (2025)
17. *Three proteins in advanced liver fibrosis: a minimalistic shallow-deep neural network approach on metabolic dysfunction-associated steatotic liver disease patients using open data*, Eleni-Myrto Trifylli, Aleksandra Leszczynska, Anastasios Kriebardis, Nikolaos Papadopoulos, Melanie Deutsch, Athanasios Angelakis (accepted EASL) (2025)
18. *Shallow-deep neural networks reveal extracellular vesicles as robust biomarkers for liver steatosis stages S0 vs. S1-S3 in metabolic dysfunction-associated steatotic liver disease patients*, Eleni-Myrto Trifylli, Athanasios Angelakis, Anastasios Kriebardis, Nikolaos Papadopoulos, Sotirios Fortis, Vasiliki Pantazatou, Ioannis Koskinas, Hariklia Kranidioti, Evangelos Koustas, Panagiotis Sarantis, Spilios Manolakopoulos, Melanie Deutsch (accepted EASL) (2025)
19. *Effect of dietary habits on outcome of nonsurgical periodontal therapy*, T.M.H. De Jong, A. Angelakis, U. Van der Velden, B.G. Loos, S. Bizzarro (submitted EuroPerio11) (2025)

20. *Extracellular Vesicles as Biomarkers for Steatosis Stages in MASLD Patients: an Algorithmic Approach Using Explainable Artificial Intelligence*, Eleni Myrto Trifylli, Athanasios Angelakis, Anastasios G Kriebardis, Nikolaos Papadopoulos, Sotirios P Fortis, Vasiliki Pantazatou, Ioannis Koskinas, Hariklia Kranidioti, Evangelos Koustas, Panagiotis Sarantis, Spilios Manolakopoulos, Melanie Deutsch - medRxiv preprint <https://doi.org/10.1101/2024.12.07.24318644> (submitted) (2024)
21. *Design and Evaluation of a CDSS for Drug Allergy Management Using LLMs and Pharmaceutical Data Integration*, G. De Vito, F. Ferrucci, A. Angelakis - arXiv preprint arXiv:2409.16395 (submitted) (2024)
22. *Evaluation of PAGE-B Score and Machine Learning Enhanced Prediction of Hepatocellular Carcinoma Development in Chronic Hepatitis B patients using Conditional Survival Forest: a multicenter study*, A. Angelakis, P. Lampertico, V. Sypsa, G. Dalekos, C. Yurdaydin, T. Berg, J. Goulis, J. L. Calleja, H. LA Janssen, M. Buti, G. Papatheodoridis, AASLD 2024 (accepted) (2024)
23. *Explainable Artificial Intelligence on Proteomics for the diagnosis of advanced liver fibrosis on MASLD patients using open data*, E. M. Trifylli, A. G. Kriebardis, N. Papadopoulos, M. Deutsch, A. Angelakis, AASLD 2024 (accepted) (2024)
24. *Effect of Daily Vitamin C Supplementation with or Without Flavonoids on Periodontal, Microbial, and Systemic Conditions Before and After Periodontal Therapy: A Case Series from an RCT*, T.M.H. De Jong, E. Stamatelou, N.A.M. Rosema, I.D.C. Jansen, B. Brandt, A. Angelakis, B. Loos, U. van der Velden, M.M. Danser, Journal of Clinical Medicine, (2024)
25. *The Prevalence of Comorbidities in Individuals with Periodontitis in a Private Periodontal Referral Practice*, N.G.F.M. Beuker, B. Loos, G. van der Heijden, E. Stamatelou, A. Angelakis, N. Su, Journal of Clinical Medicine, (2024)
26. *High-throughput transcriptome profiling indicates ribosomal RNAs to be associated with resistance to immunotherapy in non-small cell lung cancer (NSCLC)*, Moutafi MK, Bates KM, Aung TN, et al. Journal for ImmunoTherapy of Cancer 2024; 12:e009039. doi: 10.1136/jitc-2024-009039 (2024)
27. *Diagnosis of advanced liver fibrosis: The synergy of open data, synthetic data generation, CatBoost, and feature engineering*, A. Angelakis, Journal of Hepatology, vol. 80 Suppl. 1 (2024)
28. *A data-centric approach to class-specific bias in image data augmentation*, A. Angelakis, A. Rass (preprint <https://arxiv.org/abs/2403.04120>) (2024)
28. *Towards the Gene Profile of Acute Myeloid Leukaemia Using Machine Learning and Blood Transcriptomics*, A. Angelakis, R. Nathoe, M. Filippakis, Transcriptomics. Preprints 2024, 2024020593. <https://doi.org/10.20944/preprints202402.0593.v1>
30. *Diagnosis of Acute Myeloid Leukaemia on Microarray Gene Expression Data Using Categorical Gradient Boosted Trees*, A. Angelakis, I. Soulioti, M. Filippakis, Heliyon, Volume 9, Issue 10, (2023), e20530, ISSN 2405-8440, <https://doi.org/10.1016/j.heliyon.2023.e20530>
31. *Open data, explainable AI, data science and conventional NITs the recipe for new machine learning diagnostic tests on MASLD*, A. Angelakis, EASL-SLD summit (2023), Book of Abstracts
32. *Using FIB-4's parameters an explainable black-box machine learning model outperforms FIB-4 index on the diagnosis of advanced fibrosis of non alcohol related fatty liver disease patients in three cohorts from China, Malaysia and India*, A. Angelakis, T. Chen, Journal of Hepatology 2023 vol. 78(S1) | S100–S101, DOI: [https://doi.org/10.1016/S0168-8278\(23\)00435-X](https://doi.org/10.1016/S0168-8278(23)00435-X)
33. *Uncovering the Gene Profile of Acute Myeloid Leukaemia Using Gradient Boosted Trees*, A. Angelakis, Romar Nathoe, M. E. Filippakis (preprint ResearchSquare: DOI: <https://doi.org/10.21203/rs.3.rs-2374064/v1>) (2022)

34. *Performance Evaluation of Yolov4 and Yolov4-Tiny for Real-time Face-mask Detection on Mobile Devices*, K. Ntzelepi, M. E. Filippakis, M. E. Poulou, A. Angelakis, International Journal of Artificial Intelligence & Applications, (2022) Vol. 13, Num. 3
35. *Diagnosis of Acute Myeloid Leukaemia Using Machine Learning*, A. Angelakis, I. Soulioti (preprint arXiv:2108.07396) (2021)
36. *Adelic point groups of elliptic curves*, with P. Stevenhagen, Acta Arithmetica 199 (2021), 221-236, DOI: 10.4064/aa171025-27-3
37. *Classification Of Benign And Malignant Breast Lesions Using Ultrasound Shear Wave Elastography Features And BI-RADS Score: a Simple Decision Tree Algorithm*, A. Angelakis, AIUM2021, Journal of Ultrasound in Medicine
38. *Cats On The Classification Of Benign And Malignant Breast Lesions Using Ultrasound Shear Wave Elastography Features And BI-RADS Score*, A. Angelakis, AIUM2021, Journal of Ultrasound in Medicine
39. *Classification of benign and malignant breast lesions using ultrasound shear wave elastography features: a non-black-box machine learning approach*, A. Angelakis et al., p. 506, ECR 2020 Book of Abstracts. Insights Imaging 11, 34 (2020), <https://doi.org/10.1186/s13244-020-00851-0>
40. *A deep learning approach to the non-alcoholic fatty liver disease binary classification problem using patient's gender and features derived from B-mode ultrasound images regarding speed of sound and echogenicity*, A. Angelakis et al., American College of Radiology (2018)
41. *A deep-learning approach to the significant liver fibrosis binary classification problem using gender, morphologic and haemodynamic measurements derived from B-mode ultrasound images*, A. Angelakis et al., p. S279, Insights Imaging (2018) 9(Suppl 1): 1. <https://doi.org/10.1007/s13244-018-0603-8>, Springer
42. *Binary Classification of Chronic Liver Disease Patients Using Deep Learning on Morphologic B-Mode and Demographic Data*, A. Angelakis et al., AIUM 2018 Annual Convention, S3, *Journal of Ultrasound in Medicine, <https://onlinelibrary.wiley.com/doi/abs/10.1002/jum.14750>, Wiley
43. *Machine Learning for Liver's Right Lobe Volume Estimation Using Measurements on Ultrasound and Computed Tomographic Images*, A. Angelakis et al., AIUM 2018 Annual Convention, S4, *
44. *Comparison of Shear Wave Elastographic Measurement Performance on Liver's Right Lobe and Left Lobe for Diagnosis of Chronic Liver Disease Using Liver Biopsy as the "Gold Standard"*, I. Gatos, A. Angelakis et al., AIUM 2018 Annual Convention, S4, *
45. *Adelic points of elliptic curves*, with P. Stevenhagen, Proceedings of the 1st mini symposium of the Roman Number Theory Association, Vol. 1, Num. 1, 7–15, IF Press, Rome 2016
46. *Universal adelic groups for imaginary quadratic number fields and Elliptic Curves*, Doctoral Thesis, Leiden 2015
47. *Imaginary quadratic fields with isomorphic abelian Galois groups*, with P. Stevenhagen, ANTS X, Proceedings of the Tenth Algorithmic Number Theory Symposium, 21–39, Open Book Ser., 1, Math. Sci. Publ., MR3207406, Berkeley, CA, 2013

PhD Students

1. *Machine learning diagnosis of periodontitis using proteomics and cohorts: an explainability artificial intelligence approach*, Elena Stamatelou, Jan 2024, Amsterdam, Netherlands (in collaboration with Bruno Loos, ACTA, Amsterdam, NL)

Master Students

1. *Diagnosis of Acute Leukaemia by Applying Machine Learning on Gene Expression*, Romar Nathoe, 2022, Den Bosch, NL
2. *Open Street Map: Data Quality Enhancement Using Machine Learning*, Sybrand Pieter Maximiliaan Wildeboer, 2022, Den Bosch, NL
3. *Breast cancer treatment planning: a U-net approach*, Renée H.M. van Erp, 2022, Den Bosch, NL
4. *Language Domain-Specific Speech Recognition with artificially synthesized speech in Greek language*, Eleftheria Vergou, 2021, Den Bosch, NL
5. *On the Efficiency of Stock Markets - Forecasting Stock Price with Machine Learning*, Yme Nicolás van der Linden, 2021, Den Bosch, NL
6. *Explainable artificial intelligence: interpretability of a deep learning business application on seed and plant datasets*, So Hyun Kim, 2021, Den Bosch, NL
7. *A Complaint Validation and Route Optimization using Deep Learning*, Jens P.A. Bongaerts, 2021, Den Bosch, NL
8. *Survival prediction of breast cancer patients using MRI with Radiomics and Deep Learning*, Yorick Gerard Leon van Zweeden, 2021, Den Bosch, NL
9. *Scalable Detection of Promotional Overlay in Product Images*, Saskia Wessels, 2021, Den Bosch, NL
10. *A Deep Learning Solution for the Classification of Transport, Engineering, and Construction Vehicles*, Kaspar Raijmann, 2021, Den Bosch, NL
11. *Using Machine Learning for Pay-per-Usage Solutions: A Use Case Study at DLL Group*, Ad Snijders, 2021, Den Bosch, NL
12. *Predicting beer demand using machine learning: a case study at a Dutch brewery*, Lars Hanegraaf, 2020, Den Bosch, NL

Talks (invited, last 3 years)

- 2025 Data Science Foundations In Biomedicine: Our World Is Not Linear, 2025-07-07, Hangzhou Normal University, Hangzhou, China
- Data Science Foundations In Biomedicine: Our World Is Not Linear, 2025-06-27, Molecular Medicine from the Laboratory to Practice – Challenges and Questions IX conference in Athens (June 26–28), organized by Institute of Molecular Medicine & Biomedical Research and American Society of Clinical Oncology (ASCO), Athens, Greece
- Feature Selection And Non-Linearity In XAI: Advancing Biomedical Knowledge Retrieval, 2025-02-10, RWTH-Aachen University Medical Clinic, Aachen, Germany
- 2024 A New Explainable ML Methodology On Feature Selection: Application On The Diagnosis Of Acute Myeloid Leukaemia, 2024-02-19, Data Science Center, University of Amsterdam, Amsterdam
- On Feature Selection: A New Explainable ML Methodology And Application On The Diagnosis Of Acute Myeloid Leukaemia, 2024-04-11, CardioAI Seminar, Amsterdam Cardiovascular Sciences, AUMC, Amsterdam
- Digital Transformation In Healthcare, Data Analysis In Healthcare, Big Data And AI, 2024-04-11, Erasmus Medical Center, Rotterdam, Amsterdam
- Data Science And AI In Health: A Diagnostics Journey, 2024-05-17, University Of Piraeus, Athens, Greece

On Feature Selection Methods: An Alternative XAI Approach And A New ML Methodology With Applications To The Biomedical Domain, 2024-05-21, University Of Athens, Athens, Greece

Data Science Foundations In Cohorts, 2024-07-03, AUMC, Amsterdam

2023 Dimensionality Reduction on Gene Expression Microarrays Using XAI and Identification of Genes, 2023-03-01, Clinical Informatics Dept., AUMC, Amsterdam, Netherlands

Data Analysis In Healthcare; Big Data and AI, Erasmus Medical Center, 2023-04-12, Rotterdam, Netherlands

Open Science and FAIR Data: Find Your Data, AUMC, 2023-04-20, Amsterdam, Netherlands

Explainable ML Model Using FIB-4's Parameters Outperforms FIB-4 On The Diagnosis Of NAFLD Patients In three Cohorts From China, Malaysia and India, 2023-06-05, Epidemiology and Data Science Dept., AUMC, Amsterdam, Netherlands

Organising

AI in Medicine, Data Science Day, 2024-10-18, AUMC, Amsterdam

Computer skills

Programming PYTHON, MATLAB, OCTAVE, GP/PARI, SAGE, MAGMA

Data Science Jupyter, Pandas, Scikit-Learn, NumPy, TensorFlow, Keras, PyTorch, HuggingFace Transformers, LangChain, Seaborn, Matplotlib, PyDicom, OpenCV, GCP, AWS, PySpark

AI Domains Multimodal AI, Generative AI, Foundation Models, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Synthetic Data Generation

Other L^AT_EX, MySQL, QGIS, Networkx, OpenProject, Asana, Jira

Personal skills

AI Specialization Multimodal AI integration, Generative AI applications in healthcare, foundation model fine-tuning, and LLM prompt engineering

Analysis Scientific Analysis, Problem Analysis, Software Design Analysis, Root Cause Analysis

Management Leadership, Project Analysis, Project Planning, Project Management

Scholarships

10.2010–12.2013 Erasmus–Mundus Scholarship

Languages

Greek (Mothertongue), English (Fluent), Dutch (Basic-A1)